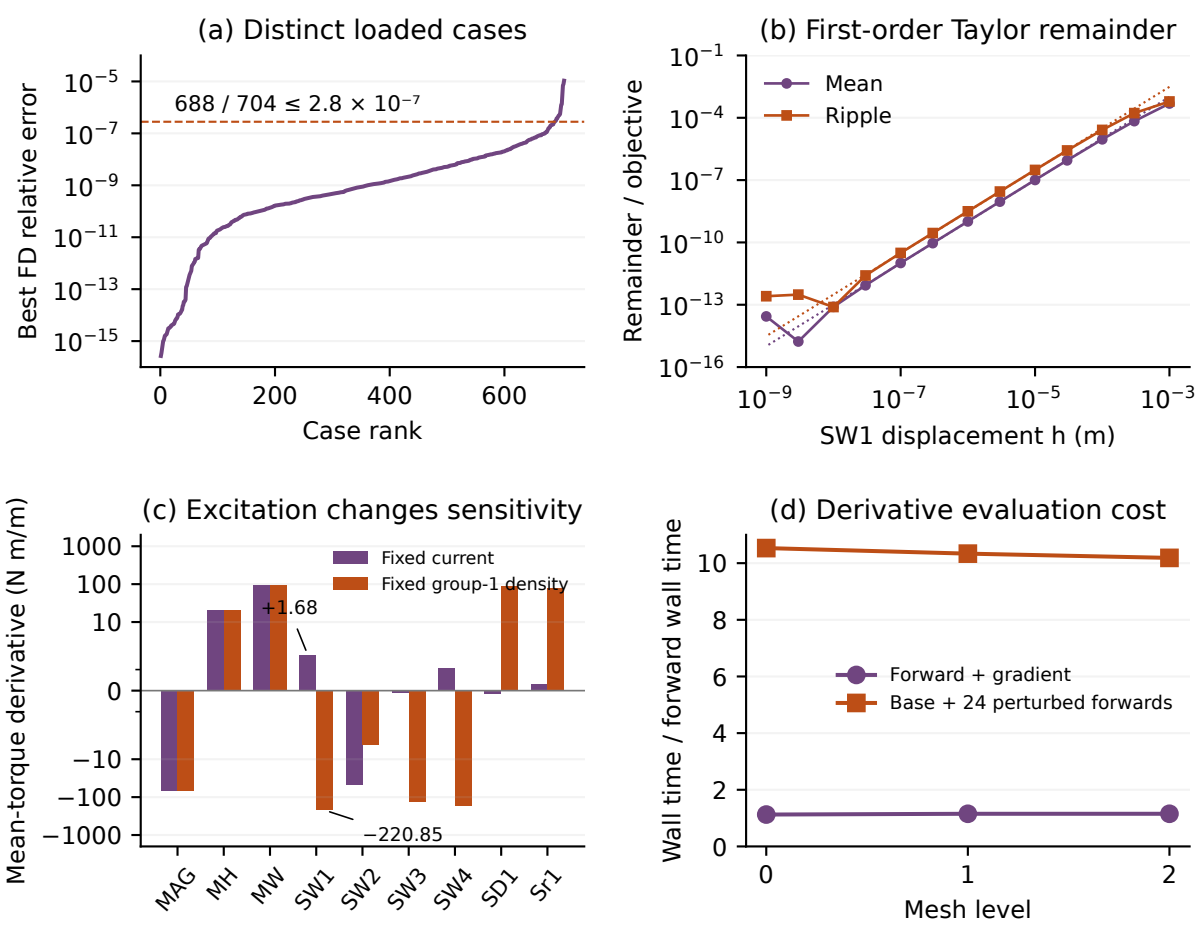
(a) Distinct loaded cases
Best FD relative error
688 / 704 ≤ 2.8 × 10⁻⁷
Case rank
(b) First-order Taylor remainder
Remainder / objective
Mean
Ripple
SW1 displacement h (m)
(c) Excitation changes sensitivity
Mean-torque derivative (N m/m)
Fixed current
Fixed group-1 density
+1.68
−220.85
MAG
MH
MW
SW1
SW2
SW3
SW4
SD1
Sr1
(d) Derivative evaluation cost
Wall time / forward wall time
Forward + gradient
Base + 24 perturbed forwards
Mesh level